



AQ11.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Choose the correct statements about $f(x, y) = x^3 + xy + y^4 - 4$.

☐

$(0, 0)$ is a critical point of f .

☐

$(0, 0)$ is the only critical point of f in $\mathbb{R}^2$.

☐

$f_{xx} = 6x$

☐

$f_{yy} = 12x^2$

☐

$f_{xy} = 1$

☐

$f_{yx} = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0)$ is a critical point of f .

$f_{xx} = 6x$

$f_{xy} = 1$

1 point

If $f(x, y) = \tan^{-1}(x/y)$, then which of the following options are correct.

☐

$f_x = \frac{y}{(x^2 + y^2)} f_x = (x^2 + y^2)y$

☐

$f_{yy} = \frac{2xy}{(x^2 + y^2)^2} f_{yy} = (x^2 + y^2)2xy$

☐

$f_{yy} = \frac{-2xy}{(x^2 + y^2)^2} f_{yy} = (x^2 + y^2)2 - 2xy$

☐

$f_{yx} = \frac{x^2 - y^2}{(x^2 + y^2)^2} f_{yx} = (x^2 + y^2)2x - y^2$

☐

$f_{yx} = \frac{y^2 - x^2}{(x^2 + y^2)^2} f_{yx} = (x^2 + y^2)2y - x^2$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_x = \frac{y}{(x^2 + y^2)} f_x = (x^2 + y^2)y$

$f_{yy} = \frac{2xy}{(x^2 + y^2)^2} f_{yy} = (x^2 + y^2)2xy$

$f_{yx} = \frac{x^2 - y^2}{(x^2 + y^2)^2} f_{yx} = (x^2 + y^2)2x - y^2$

1 point



If $f(x, y) = x^2 - 3xy^3$, then the determinant of the Hessian matrix $Hf(x, y)$ is

☐

$36xy + 81y^4$

☐

00

☐

$-36xy - 81y^4$

☐

$36xy - 81y^4$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$-36xy - 81y^4$

Level 2

1 point

Choose the correct statements about $f(x, y) = x^2 + y^4$.

☐

$(0, 0)$ is a critical point of f .

☐

Rank of the Hessian matrix $Hf(0, 0)$ is 22.

☐

The determinant of the Hessian matrix $Hf(0, 0)$ is 00.

☐

The determinant of the Hessian matrix $Hf(0, 0)$ is 11.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0)$ is a critical point of f .

The determinant of the Hessian matrix $Hf(0, 0)$ is 00.

Number of critical points of the function $f(x, y) = x^3 + y^3 - 3x - 3y$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

1 point

For which of these functions are the mixed partial derivatives the same at $(0, 0)$.

☐

$f(x, y) = 3x^2 + 7y^3 - 2x^2y^3$

☐

$f(x, y) = \begin{cases} 1 & y \neq 0 \\ 0 & y = 0 \end{cases}$

☐

$f(x, y) = |x|(y^2 + 1)$

☐

$f(x, y) = ax + by$, where a and b are two fixed real numbers.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(x, y) = 3x^2 + 7y^3 - 2x^2y^3 \quad f(x, y) = 3x^2 + 7y^3 - 2x^2y^3.$$

$$f(x, y) = ax + by \quad f(x, y) = ax + by, \text{ where } a \text{ and } b \text{ are two fixed real numbers.}$$

[Check Answers](#)

Your score is: 0/6